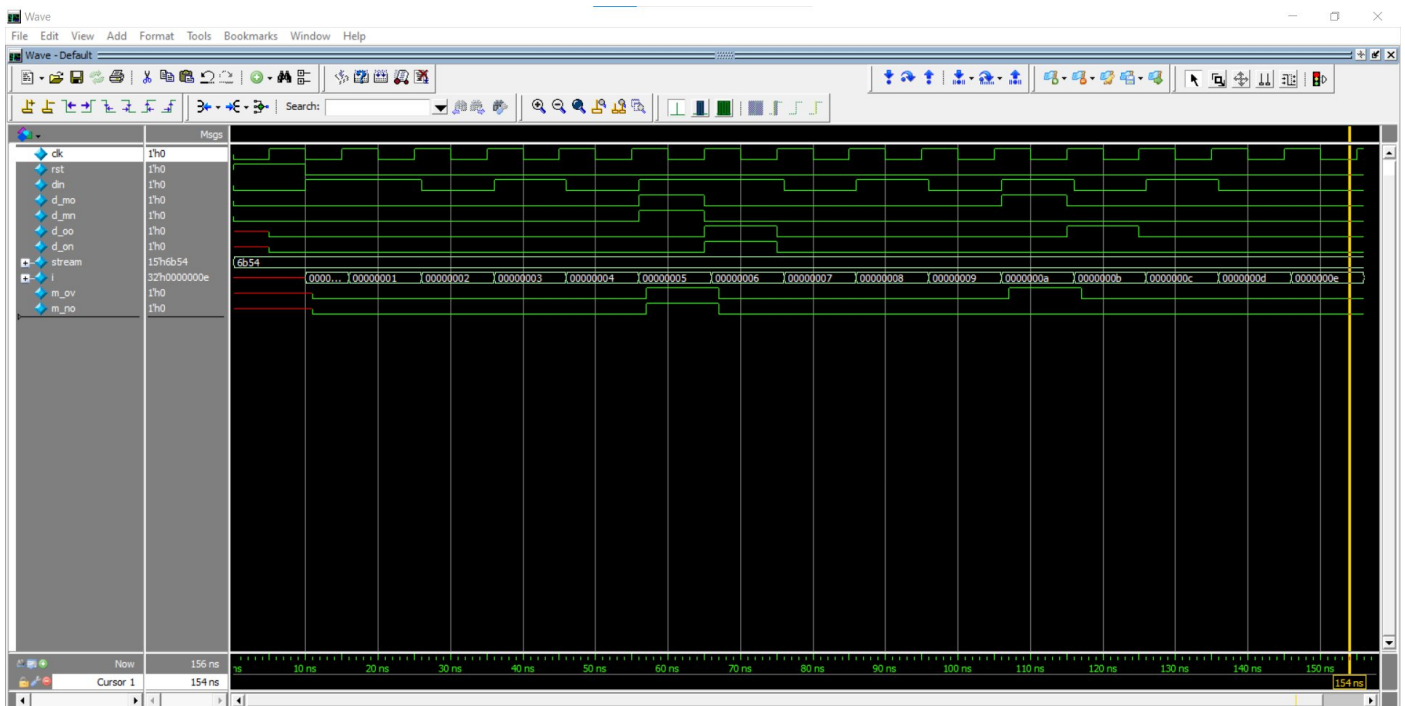


Sequence detector

Waveform snapshot:



#	t	bit	MealyOv	MealyNo	MooreOv	MooreNo
#	0	1	0	0	0	0
#	1	1	0	0	0	0
#	2	0	0	0	0	0
#	3	1	0	0	0	0
#	4	0	0	0	0	0
#	5	1	1	1	1	1
#	6	1	0	0	0	0
#	7	0	0	0	0	0
#	8	1	0	0	0	0
#	9	0	0	0	0	0
#	10	1	1	0	1	0
#	11	0	0	0	0	0
#	12	1	0	0	0	0
#	13	0	0	0	0	0
#	14	0	0	0	0	0
#	** Note: \$finish : C:/Users/hp/Desktop/Digital design					

Results

All four detector variants (Mealy/Moore $\times$ overlapping/non-overlapping) were simulated against the test stream 110101 101010 1.

- **t=5:** all four detectors correctly fire after the first complete 110101.
- **t=10:** only the **overlapping** Mealy and Moore detectors fire again; the **non-overlapping** ones stay at 0. This is expected — the stream contains a second, overlapping occurrence of the pattern, which only the overlapping designs are built to catch (they reuse the bits instead of resetting).

Conclusion: simulation matches the theoretical state tables exactly. Mealy and Moore give identical detection timing for the same policy (overlap vs. non-overlap) — they only differ in *how* the output is generated. The two policies only diverge in behavior when the input actually contains an overlapping repeat of the target sequence.